G-DiffPS: Physics-Informed Graph Diffusion Policy for Amortized Multi-Topology RF Phase Shifter Synthesis

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Abstract

Designing RF phase shifters at millimeter-wave (mmWave) frequencies requires iterative SPICE simulation loops that consume hours of compute per specification target. We present G-DiffPS, to our knowledge the first graph-conditioned *conditional flow matching* (CFM) policy for RF phase-shifter synthesis. Once trained, G-DiffPS emits SPICE-ready component values in ≈2 ms per specification across the 1–40 GHz band, at an inference cost that does not grow with the topology or the frequency target. Each in-distribution design is specification-compliant on emission—99% first-pass SPICE yield, and a passing design on nine of ten held-out scenarios, so the per-specification SPICE search that Differential Evolution, Simulated Annealing, and Bayesian Optimization repeat for every target is paid once, at training time, not at every deployment. We claim amortization and topology transfer, not higher converged quality than these optimizers. Three mechanisms make this possible: (i) a *physics-informed logarithmic action warping* that resolves the mmWave "needle-in-a-haystack" search bottleneck, yielding a 465× first-pass yield improvement on the Loaded-Line topology; (ii) a *Graph Isomorphism Network (GIN) topology encoder* that conditions the actor on heterogeneous circuit graphs under one set of weights; and (iii) a *dual-tier ABCD-matrix pre-filter* that rejects ≈80% of non-resonant candidates before SPICE, cutting training simulation overhead 4.7×. Leave-one-out cross-validation isolates where graph conditioning transfers: four of six held-out topologies reach 61–100% zero-shot compliance, and the two failures trace to a training/target frequency-band mismatch: a distributional limit we analyze directly rather than to encoder capacity. Code and artifacts are available here https://github.com/ACADLab/G-DiffPS

CCS Concepts

• **Computing methodologies → Reinforcement learning**; • **Hardware → Electronic design automation**.

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1 Introduction

RF phase shifters anchor 5G/6G beamforming arrays, phased-array radars, and millimeter-wave transceivers. Designing one to a target S-parameter specification means searching a high-dimensional, nonlinear space of component values under simultaneous constraints on return loss (S_{11}), insertion loss (S_{21}), and phase coverage. At mmWave the search turns hostile: component values carry sub-pF precision targets, and resonances sharpen until a single 0.1 pF step crosses clean over the resonant peak.

Classical optimizers absorb this cost once per specification and then discard it. Bayesian Optimization (BO) [2], Differential Evolution (DE), Simulated Annealing (SA), and Nelder-Mead simplex search each restart from scratch on a new target, typically 10^2–10^4 SPICE evaluations deep [2]. Graph Neural Network (GNN)-augmented reinforcement learning [3, 13] transfers sizing across topologies but still searches at inference. The limitation is structural, not algorithmic: each method models a *single* objective landscape and rebuilds it for every new specification. None reuses the effort spent on the last one.

We propose G-DiffPS, which amortizes that search instead of repeating it. In one offline training phase, G-DiffPS learns a continuous, specification-conditioned map from a 12-dimensional target to physical component values. At deployment a 10-step deterministic Ordinary Differential Equation (ODE) solve returns a complete design in ≈2 ms with no SPICE calls. The inference cost is constant: it does not grow with the topology or the frequency target. Design *quality*, by contrast, still depends on whether the target lies inside the trained distribution—a boundary we characterize directly in Section 5.3. Three tightly coupled mechanisms make this possible, each serving a claim the evaluation then tests:

(1) Physics-Informed Log Warping. We derive a logarithmic action space aligned with LC resonance physics, yielding a 465× first-pass yield improvement on the Loaded-Line topology (Section 4). This is what lets a first-pass design clear specification at zero search. **(2) Graph-Conditioned CFM Policy.** A Graph Isomorphism Network (GIN) topology encoder with sum-pooling conditions a conditional flow matching actor [6] that handles all six heterogeneous phase-shifter topologies under one set of weights. Leave-one-out cross-validation quantifies zero-shot transfer to unseen topologies (Section 5.3); we further show the zero-shot ceiling is set by the actor's learned parameter prior, not the encoder (Appendix A). (3) **Dual-Tier Simulation Shield.** An analytical ABCD-matrix pre-filter in PyTorch rejects ≈80% of non-resonant candidates in microseconds, cutting training SPICE overhead 4.7× with < 0.02° quality impact (Section 3.3)—what keeps the online training loop tractable.

We carry three claims through the evaluation, labeled **C1–C3**. **C1 (Amortization):** synthesis costs zero inference-time SPICE against the per-specification search optimizers repeat for every target (Section 5.2). **C2 (Zero-cost compliance):** the emitted design is specification-compliant on arrival—99% first-pass yield in distribution (Section 5.1). **C3 (Zero-shot transfer):** the graph-conditioned actor transfers to unseen topology graphs (Section 5.3). We claim amortization and transfer, not higher converged quality than the optimizers we compare against.

2 Related Work

Classical optimization. Bayesian Optimization with Gaussian-process surrogates is the standard black-box approach to analog and RF sizing [2]: it is sample-efficient on a single specification because the surrogate concentrates sampling near promising regions, but the surrogate is discarded once the target changes. Differential Evolution and Simulated Annealing reach competitive designs without a surrogate, at the cost of more evaluations [9]. The common limitation is structural, not algorithmic: each method models a *single* objective landscape and must rebuild that model from scratch for every new specification. G-DiffPS instead fits one specification-conditioned map offline and queries it in constant time, so the optimization cost is paid once rather than per target.

GNN-augmented RL for circuit sizing. GCN-RL [13] encodes transistor topology graphs and transfers sizing across technology nodes through reinforcement learning; GCX [3] adds semi-supervised graph optimization for few-shot sizing. Both establish that a graph encoder can carry structural knowledge between circuits, and we build on that premise. Neither targets RF S-parameter objectives, and more to the point of this paper, both still run an iterative policy search at inference, so the deployment-time cost they incur is the cost we remove.

Diffusion and flow matching for control. Chi et al. [1] established diffusion policies for multimodal robotic control, showing that a generative actor captures the several distinct modes a unimodal Gaussian policy collapses. Conditional flow matching [6] reaches the same expressiveness without stochastic sampling: a vector field v_θ regresses straight-line paths between noise and data, which yields deterministic single-trajectory inference and a gradient signal of uniform magnitude across the interpolation parameter. For RF sizing both properties earn their place, determinism makes a synthesized design reproducible, and the uniform gradient stabilizes the online loop where the reward arrives from a noisy SPICE evaluation. Score-based models [8] and flow matching have been applied to molecular generation; we adapt them to a setting they have not previously addressed, where the action space demands a physics-derived warp and the reward is a circuit simulation.

LLMs for circuit design automation. A complementary line of work prompts large language models (LLMs) to generate Hardware codes [12], specifically SPICE code and analog designs directly: SPICEPilot [10] guides LLMs to emit simulation-ready netlists, and LLM-driven frameworks automate analog in-memory architecture exploration [11]. These approaches rely on the LLM's pretrained priors and lack a SPICE-in-the-loop reward or a physics-warped continuous action space; we include an LLM-as-sizer baseline (Appendix D) and find it trails G-DiffPS on resonance-sensitive mmWave specifications.

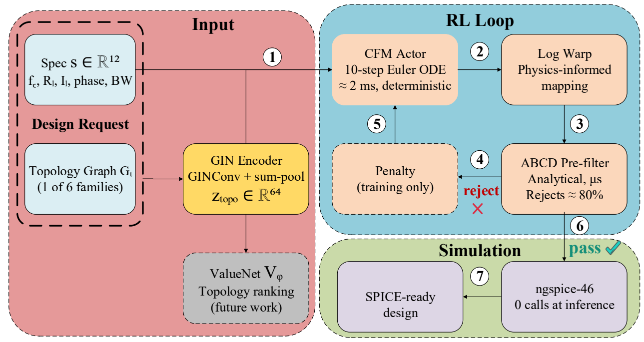

Figure 1: G-DiffPS framework pipeline. ① Design request: specification $\mathbf{s} \in \mathbb{R}^{12}$ and topology graph $\mathcal{G}_T$. ② GIN encoder: GINConv + sum-pool → $\mathbf{z}_{\text{topo}} \in \mathbb{R}^{64}$. ③ CFM actor: 10-step Euler ODE solve, ≈2 ms, deterministic. ④ Log warp: physics-derived base-10 exponential mapping to physical component values. ⑤ ABCD pre-filter: analytical S-parameter feasibility check, rejects ≈80% in microseconds. ⑥ `ngspice-46`: full simulation during training only; zero SPICE calls at inference. ⑦ SPICE-ready design output. Dashed modules are training-only or deferred extensions.

3 The G-DiffPS Framework

Figure 1 traces the full G-DiffPS pipeline through seven numbered stages. **Step** ① presents the design request: a 12-dimensional specification **s** encoding center frequency f_c, return loss, insertion loss, phase coverage, and bandwidth, paired with a topology graph $\mathcal{G}_T$ drawn from one of the six supported phase-shifter families. **Step** ② encodes the topology: a three-layer GIN maps $\mathcal{G}_T$ to a 64-dimensional structural code $\mathbf{z}_{\text{topo}} \in \mathbb{R}^{64}$ via GINConv layers with sum-pooling (Section 3.1). **Step** ③ runs the Conditional Flow Matching (CFM) actor: conditioned jointly on **s** and $\mathbf{z}_{\text{topo}}$, a 10-step deterministic Euler ODE solve produces a normalized action vector $\hat{\mathbf{x}}$ in ≈2 ms (Section 3.2). **Step** ④ applies the physics-informed log warp: $\hat{\mathbf{x}}$ is mapped to physical component values **a** via the base-10 exponential scaling derived from LC resonance physics (Section 4). **Step** ⑤ screens **a** through the dual-tier ABCD-matrix pre-filter: candidates failing basic resonance compliance are rejected analytically in microseconds, without a SPICE call, discarding ≈80% of proposals during training. **Step** ⑥ commits passing candidates to `ngspice-46` during training; at deployment this stage is bypassed—zero SPICE calls at inference. **Step** ⑦ delivers the SPICE-ready design. The 4 core stages (②–⑤) execute once in sequence at deployment with no search. Dashed modules (the penalty signal and value head V_ψ) are training-only or deferred extensions.

3.1 GNN Topology Encoder

Each of the six phase-shifter topologies (*Loaded Line*, *Switched Line*, *Reflection Type*, *Switched Filter*, *Vector Modulator*, *All Pass*) is a PyTorch Geometric graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with 5-dimensional one-hot node features encoding component class (`TLine`, `Series_Switch`, `Shunt_Cap`, `Inductor`, `Port`).

We use a three-layer Graph Isomorphism Network (GIN) [14] with injected node degree, node-level LayerNorm, and *sum* readout:

$$\mathbf{z}_{\text{topo}} = W\left(\textstyle\sum_{v\in\mathcal{V}} \text{GIN}(\mathcal{G})_v\right) \in \mathbb{R}^{64}. \quad (1)$$

Both design choices are deliberate for small, dense RF graphs. *Sum* aggregation (vs. mean) preserves neighbor count on one-hot inputs, and *sum* pooling (vs. mean pooling) preserves graph size—together they give the maximal discriminative power of the 1-Weisfeiler–Leman (1-WL) graph isomorphism test [14]. Mean-pooled architectures (e.g. GraphSAGE) instead collapse all six topology embeddings onto a single hypersphere (pairwise $L_2 \approx 0.06$), rendering the encoder topology-blind; GIN restores structurally distinct embeddings (pairwise $L_2 \approx 4.0$). We analyze this representation collapse and its (notably bounded) effect on downstream transfer in Appendix A.

3.2 Conditional Flow Matching Sizing Policy

The sizing policy is a conditional flow matching (CFM) model conditioned jointly on $\mathbf{z}_{\text{topo}}$ and the 12-dimensional target specification $\mathbf{s} \in \mathbb{R}^{12}$ (center frequency, return loss, insertion loss, phase coverage, bandwidth).

During training, we define a straight-line path from noise $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ to data $\mathbf{x}_0$:

$$\mathbf{x}_t = (1 - t)\,\boldsymbol{\epsilon} + t\,\mathbf{x}_0, \quad t \in [0, 1]. \tag{2}$$

A vector field network v_θ is trained to predict the conditional flow velocity:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t,\mathbf{x}_0,\boldsymbol{\epsilon}}\left[w(\mathbf{x}_0)\left\|v_\theta(\mathbf{x}_t,\ t,\ \mathbf{s},\ \mathbf{z}_{\text{topo}}) - (\mathbf{x}_0 - \boldsymbol{\epsilon})\right\|^2\right], \tag{3}$$

where $w(\mathbf{x}_0) = \text{clamp}(\exp(A/\tau_w), 10)$ is an advantage weight ($A = Q_\phi - V_\psi$, temperature $\tau_w = 0.5$) that up-weights high-reward component configurations during online training. At inference, a 10-step Euler ODE solve $\mathbf{x}_{t+\Delta t} = \mathbf{x}_t + \Delta t \cdot v_\theta(\mathbf{x}_t, t, \mathbf{s}, \mathbf{z}_{\text{topo}})$ produces the same output for the same input—a deterministic read-off from the learned synthesis manifold.

The vector field v_θ is a multilayer perceptron that consumes the concatenation of four inputs: the current state $\mathbf{x}_t \in \mathbb{R}^9$, a sinusoidal embedding of the flow time t, the normalized specification $\mathbf{s}$, and the topology code $\mathbf{z}_{\text{topo}}$. The nine-dimensional action covers the largest parameter set across the six topologies; for topologies with fewer free components the unused coordinates are masked before warping, so a single actor head serves every graph. Conditioning on $\mathbf{z}_{\text{topo}}$ rather than a one-hot topology index is what makes the policy extend to unseen graphs: the actor reads structure, not identity, so a held-out topology that shares sub-graph motifs with the training set inherits a usable parameter distribution (Section 5.3). We fix the solver at ten Euler steps; below eight steps the discretization error of the straight-line flow begins to perturb the sub-pF coordinates, and beyond ten the design is unchanged while latency grows, so ten balances fidelity against the ≈2 ms inference target.

3.3 Dual-Tier Simulation Shield

SPICE simulation dominates compute cost. We interpose an analytical ABCD-matrix pre-filter [7] implemented in PyTorch that estimates S-parameter feasibility in microseconds. Candidates failing basic resonance compliance are rejected before `ngspice-46`. This pre-filter rejects ≈80% of candidates, cutting total SPICE calls from 10,000 to 2,143 over a 10,000-step training run—a 4.7× overhead reduction with <0.02° quality impact (Table 3).

Algorithm 1: G-DiffPS online advantage-weighted training

```
Input: topology graphs {G_1, ..., G_6}, spec sampler P_s, budget N
Output: CFM actor v_θ, GIN encoder f_G
Initialize v_θ, critic Q_φ, value net V_ψ, encoder f_G
for step = 1 to N do
    sample topology T, spec s ~ P_s(T)
    z ← f_G(G_T);   ε ~ N(0, I)
    x̂ ← EulerSolve(v_θ, z, s, ε, 10)
    a ← LogWarp(x̂)                                    // Eq. 4
    if not ABCD_PASS(a, s, T) then
        r ← r_penalty; continue
    r ← Spice(a, s, T)
    A ← Q_φ(x̂, s, z) − V_ψ(s, z);   w ← clamp(e^{A/τ_w}, 10)
    update v_θ on w‖v_θ(x_t, t, s, z) − (x_0 − ε)‖²
    update Q_φ, V_ψ on IQL expectile loss (τ=0.7)
end
```

3.4 Online Advantage-Weighted Training

G-DiffPS requires *zero pre-existing datasets*; it generates its own supervision online. Algorithm 1 states the loop. At each step the actor proposes component values for a sampled topology, the log warp maps them to physical units, the ABCD filter screens them, and `ngspice-46` returns a reward. That reward drives two updates. The advantage $A = Q_\phi - V_\psi$ weights the flow-matching regression so the actor moves toward high-reward configurations; the critic Q_ϕ and value network V_ψ fit the reward through Implicit Q-Learning (IQL) expectile regression ($\tau = 0.7$) [4], which isolates the high-performance frontier of the reward distribution without multi-step rollouts. The clamp on w caps the influence of any single lucky sample, keeping the regression target bounded on the rare steps where SPICE reports an outlier reward. The loop sustains ≈6.9 steps/second on a single NVIDIA A100 80 GB GPU, so a 10,000-step run completes in under half an hour.

4 Physics-Informed Logarithmic Action Warping

4.1 The Linear Scaling Failure at mmWave

A standard continuous policy maps its output $a \in [0, 1]$ to component values linearly: $C = a(C_{\max} - C_{\min}) + C_{\min}$. For a 28 GHz Loaded-Line phase shifter, LC resonance requires $C_{\text{load}} \approx 0.05$ pF. With bounds $[0.005, 10.0]$ pF, $a = 0.5$ yields 5.0 pF—two orders of magnitude above the resonant target, and a $\Delta a = 0.01$ exploration step shifts C by 0.1 pF, jumping entirely over the resonant sweet spot. The Loaded-Line topology achieved only 0.2% first-pass yield under linear scaling.

4.2 Logarithmic Manifold Warping

We replace the linear map with a base-10 exponential mapping:

$$C = 10^{\,a\cdot\log_{10}(C_{\max}/C_{\min})+\log_{10}(C_{\min})}. \tag{4}$$

With $[C_{\min}, C_{\max}] = [0.005, 0.5]$ pF, $a = 0.5$ maps precisely to $10^{0.5\times 2-2.301} = 0.050$ pF—the resonant target. This is a physical derivation, not a hyperparameter: mmWave resonances scale as $f_0 = 1/2\pi\sqrt{LC}$, an inherently logarithmic relationship. A unit step in log-space corresponds to a constant *fractional* change in component value—the correct inductive bias for resonance search.

Table 1: Linear vs. log action space: first-pass yield across all six topologies (3 seeds, 10,000 steps).

Topology	Linear	Log	Gain
Loaded Line	0.2%	**93.0%**	465×
Reflection Type	1.1%	**100.0%**	91×
Switched Filter	43.5%	**100.0%**	2.3×
All Pass	76.8%	**100.0%**	1.3×
Vector Modulator	84.2%	**100.0%**	1.2×
Switched Line	100.0%	**100.0%**	—

Under linear scaling the reward landscape is, for the policy, effectively flat: the resonant peak occupies a Δa window narrower than a single exploration step, so it is invisible to gradient and to random sampling alike. Log warping recenters the action distribution onto that peak, turning an unsearchable target into the median of the sampling range. The effect is quantified per topology in Table 1.

4.3 Topology-Dependent Warping Analysis

The warp helps unevenly, and the pattern is itself diagnostic. Table 1 reports first-pass yield under linear and log action spaces for all six topologies. Resonance-sensitive structures separate sharply from tolerant ones. Loaded Line and Reflection Type place a sub-pF capacitor at the heart of a sharp LC resonance, so linear scaling—which spends half its dynamic range above 5 pF—almost never lands inside the resonant window, and first-pass yield collapses to 0.2% and 1.1%. Log warping lifts both past 93%, a 465× gain on Loaded Line. Switched Filter, All Pass, and Vector Modulator carry broader component tolerances; they converge under either map, and the warp contributes a modest 1.2×–2.3×. Switched Line, whose phase shift comes from transmission-line *length* rather than a tuned resonance, is indifferent to the warp and reaches full yield in both spaces. The warp is therefore not a global trick but a targeted correction: it matters exactly where the physics is logarithmic, and it is inert where the physics is not. This is why we derive its bounds per topology rather than tuning a single global scale, and why the two topologies still lacking analytic bounds (Switched Filter, All Pass) are the ones whose joint-training yield trails in Table 2.

5 Experimental Evaluation

All experiments use a single NVIDIA A100 80 GB GPU, AMD EPYC 7763 CPU, custom-compiled `ngspice-46`, and PyTorch 2.3 / PyTorch Geometric 2.5.3. SPICE netlists use realistic switch parasitics ($R_{\text{on}} = 6.5\,\Omega$, $C_{\text{off}} = 650\,\text{fF}$). All results are mean ± std over 3 seeds (42, 1337, 2026). The scalar reward aggregates the S-parameter objectives that define a phase-shifter specification. For each design we run every phase state in `ngspice-46`, then score the worst-case return loss S_{11}, the insertion loss S_{21}, and the RMS phase error against the target phase set, each as a margin relative to its specification bound and clipped so that a comfortably-met constraint does not mask a violated one. A design that satisfies every constraint scores positively; a design that the physics pre-filter rejects, or whose simulation diverges, receives a fixed penalty. We treat reward > 0.5 as spec compliance throughout—the threshold at which all three margins are met with headroom. "Yield" counts proposals that pass SPICE at all; "compliance" counts those that clear the 0.5 bar.

Table 2: G-DiffPS convergence: a single unified CFM actor (GIN encoder) trained jointly on all 6 topologies (10,000 steps; converged statistics over the final 4,000 steps, seeds 42/1337). Yield is the fraction of proposals passing SPICE; reward is unified SPICE quality score that the framework optimizes.

Topology	Yield	Best Reward	Mean Reward
Switched Line	100.0%	2.26	0.88
Reflection Type	100.0%	2.23	0.81
Vector Modulator	100.0%	2.14	0.69
Loaded Line	99.3%	1.82	0.65
Switched Filter	82.0%	2.10	0.73
All Pass	49.0%	0.96	0.57

Table 3: Component ablation by SPICE cost/feasibility (Loaded Line yield; GIN encoder held fixed across rows). "Inference SPICE" is the per-deployment cost that C1 targets.

Configuration	LL Yield	Infer. SPICE	Train SPICE
Full G-DiffPS (CFM)	**99.3%**	**0**	**2,143**
− ABCD pre-filter	≈99%	0	10,000
− Log warp (linear)	0.2% (div.)	0	≈9,800
CFM → SAC (online RL)	—	10	10,000

5.1 Multi-Topology Convergence and Zero-Cost Compliance (C2)

Table 2 reports the converged performance of a single unified G-DiffPS actor trained jointly on all six topologies under the GIN encoder (Section 3.1). All methods reported in this paper use this same GIN encoder; no results mix encoder variants. Four of six topologies reach ≥ 99% SPICE yield with best rewards of 2.1–2.3; the two lower-yield topologies (Switched Filter, All Pass) are precisely those whose log-warp bounds have not yet been derived analytically (Section 4), isolating the remaining gap to a known, addressable cause rather than the generative policy itself. Each design emerges specification-compliant at zero inference-time SPICE cost (C2): the amortized actor reaches first compliance without the per-specification search that iterative optimizers repeat. We make no claim of higher converged reward than Differential Evolution or Simulated Annealing—Section 5.2 shows those methods keep improving with budget—but G-DiffPS removes that budget from deployment. Table 3 isolates each contribution by its effect on SPICE cost and feasibility. The ABCD pre-filter is the dominant efficiency lever: it rejects ≈80% of candidates analytically, cutting the training SPICE budget from 10,000 to 2,143 calls (4.7×) at unchanged yield. Log warping is the dominant *feasibility* lever: without it the linear action space collapses to 0.2% first-pass yield on Loaded Line and fails to converge on resonance-sensitive topologies (Section 4). The online SAC baseline (with the same GIN encoder) reaches comparable reward but only by spending 10 SPICE calls *per deployment*, whereas the CFM actor spends none—the core amortization distinction (C1).

5.2 Amortized Deployment vs. Iterative Search (C1)

We measure the amortization claim on ten representative scenarios (Table 4) that span all six topologies and four frequency bands, from a 2.4 GHz All-Pass network to a 38 GHz Loaded-Line shifter. The comparison separates two quantities that the literature often

Table 4: The 10 deployment scenarios: topology, center frequency, phase coverage, and minimum return-loss target. Together they span the 1–40 GHz and all 6 families.

ID	Topology	f_c (GHz)	Cov. (°)	S_{11} (dB)
S1	Loaded Line	28	180	18
S2	Loaded Line	38	180	15
S3	Switched Line	14	180	15
S4	Switched Line	24	180	12
S5	Vector Modulator	5	360	20
S6	Vector Modulator	8	360	18
S7	Reflection Type	28	90	12
S8	Reflection Type	10	180	15
S9	All Pass	2.4	180	18
S10	Switched Filter	18	90	12

conflates: the cost paid *per deployment* and the quality reachable *at a given budget.*

Per-deployment SPICE cost. G-DiffPS performs *zero* SPICE calls at inference: the design is the output of a single ODE solve. Every iterative baseline—DE, SA, BO, Nelder-Mead, and online SAC—instead issues a fresh stream of SPICE evaluations for each scenario, because each restarts its search from an uninformed state. Over a deployment of K specifications the asymmetry compounds: G-DiffPS amortizes one training run across all K, while the baselines pay their full per-scenario budget K times over.

Quality at deployment budget B. Figure 2 plots the best reward each method reaches as a function of the SPICE budget $B \in \{1, 10, 50, 200, 1000\}$. G-DiffPS appears as a horizontal line at $B = 0$: its quality is fixed because it consults no simulator. The iterative methods start below that line and cross it within a handful of calls—they reach first spec compliance at $B \approx 1$–10 and continue climbing to higher reward with more budget. We read this honestly. G-DiffPS does not win the asymptote; granted enough simulation, DE and SA refine designs past what the amortized actor produces. What G-DiffPS wins is the first compliant design at zero cost, which is exactly the regime that dominates large design sweeps and warm-starting. Used as a warm start, the amortized design hands each optimizer a feasible initialization and removes the $B \approx 1$–10 calls it would otherwise spend reaching feasibility—a saving that multiplies across every specification in a sweep.

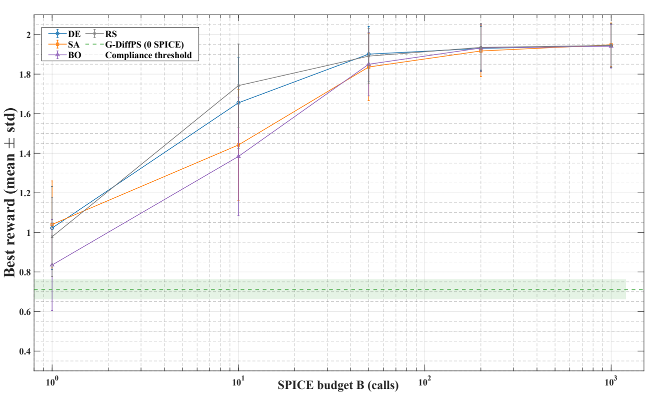

Figure 2: Best reward vs. SPICE budget B across 10 paper scenarios (mean ± std, 3 seeds). G-DiffPS (zero SPICE at inference, dashed) reaches first spec compliance at zero SPICE; baselines need $B \approx 1$–10 calls.

Table 5: Zero-shot LOOCV per held-out topology (GIN encoder, 200 samples, mean over 3 seeds). The compliance/prior-pass *gap* is shown visually in Figure 3; here we give the exact compliance and the converged best reward.

Held-out topology	Compliance	Best reward
Switched Line	100.0%	2.17
Vector Modulator	100.0%	2.17
Loaded Line	76.7%	2.04
Reflection Type	61.2%	2.10
Switched Filter	17.7%	1.55
All Pass	0.0%	−3.18

5.3 Zero-Shot Graph Transfer via LOOCV (C3)

To evaluate whether the GIN encoder enables transfer of structural physics, we conduct a Leave-One-Out Cross-Validation (LOOCV) experiment: G-DiffPS is trained for 5,000 steps on five topologies with each topology held out in turn. At test time—*zero gradient updates on the held-out graph*—the GIN-conditioned actor receives only the unseen topology's graph structure and a fresh specification, and we sample 200 designs.

Table 5 reports zero-shot compliance for all six holdout conditions (GIN encoder, 3 seeds). Four of six topologies transfer at ≥ 61%: Switched Line and Vector Modulator reach 100%, while Loaded Line (76.7%) and Reflection Type (61.2%) retain strong compliance despite never being seen during training. We report the two failure cases plainly. Switched Filter (17.7%) clears the physics pre-filter for 65% of samples but meets the full S-parameter spec for only 17.7%. All Pass (0%) fails outright: its 2.4 GHz target lies an order of magnitude below the 10–28 GHz training band, so the actor's learned LC parameter prior is physically mismatched to All Pass's resonance scale—a distributional, not architectural, limitation.

Figure 3 separates the two rates that the single compliance number conflates. For Switched Line and Vector Modulator the physics-prior pass rate and the full-spec compliance coincide at 100%: every emitted design is both structurally sane and spec-meeting. Switched Filter exposes the gap that matters for diagnosis—65% of its samples pass the analytical prior, yet only 17.7% clear the full S-parameter target, so the actor places the design in the right region of the space but does not land the precise resonance the held-out filter demands.

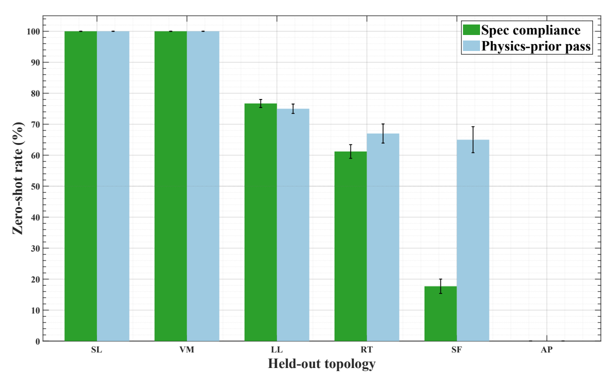

Figure 3: Zero-shot LOOCV per held-out topology (GIN encoder, 3 seeds): full spec compliance (green) and physics-prior pass rate (blue). The gap between the two—widest for Switched Filter—marks designs that are structurally plausible but not yet spec-precise.

The transfer pattern tracks structural overlap rather than electrical similarity: Switched Line shares its series-switch motif with several training graphs and transfers perfectly, whereas All Pass, whose bridged-T topology and sub-2.5 GHz band are unique in the set, has no structural neighbor to borrow from. Graph conditioning transfers what the training topologies share; it cannot invent a parameter regime the actor has never been rewarded in.

Crucially, the GIN encoder is *not* the bottleneck for the two failure cases. An encoder ablation (Appendix A) shows that the LOOCV compliance is statistically invariant across a collapsed mean-pooled encoder ($L_2 \approx 0.06$) and our structurally distinct GIN encoder ($L_2 \approx 4.0$): the actor's cross-topology parameter prior, not the embedding geometry, sets the zero-shot ceiling. This is the honest and reproducible account of where graph conditioning helps and where it does not.

5.4 Why Flow Matching, Not Diffusion

The choice of a flow-matching actor over a Denoising Diffusion Probabilistic Model (DDPM) actor is deliberate; Appendix B (Figure 4) isolates its effect under matched conditions. We train both under identical conditions—the same GIN encoder, log sizing, IQL critic, and three seeds—changing only the generative head. Two differences matter at our scale. Flow matching regresses a straight-line velocity field with uniform weight across the interpolation parameter t, whereas the DDPM objective re-weights timesteps through a noise schedule; on the sparse, low-dimensional action space of phase-shifter sizing the uniform signal converges faster and with lower seed-to-seed variance. At inference the gap widens into a practical one: flow matching reads off a design in a 10-step deterministic ODE solve, while DDPM requires many stochastic denoising steps for comparable quality. Both actors reach the same reward plateau, and both reach it by ≈2,500 steps—the 10,000-step budget used throughout the paper therefore carries more than a threefold margin (Appendix B).

6 Discussion

Why amortization is the right frame. A single converged DE or SA run can outscore G-DiffPS on one specification. The contribution is not a better optimizer—it is the removal of the optimizer from the deployment loop. Design sweeps make this concrete: a phased-array campaign sizes hundreds of element variants across frequency, bias, and process corners; an iterative method repeats its full search for each, while G-DiffPS answers every query in a constant-cost forward pass. The break-even point is low. Because the amortized design already clears spec compliance, the appropriate baseline comparison is not asymptotic reward but calls-to-first-feasible—G-DiffPS reports zero where iterative methods report $B \approx 1$–10 per specification (Section 5.2). **Differentiable inference.** G-DiffPS is differentiable end to end. Because the entire pipeline is expressed in PyTorch, the Jacobian $\partial \mathbf{a}_0 / \partial \mathbf{s}$ of the emitted design with respect to the specification is available by automatic differentiation. That opens gradient-based sensitivity analysis and secondary-objective refinement—trimming power while holding phase error fixed, for instance—that no black-box SPICE optimizer exposes. **Scope and limitations.** Log-warp bounds are derived analytically for Loaded Line only; extending them to Reflection Type and the remaining four topologies is the immediate next step, expected to lift the two low-compliance LOOCV folds. G-DiffPS synthesizes parameters for a *given* topology: automatic topology recommendation remains open—the value head V_ψ, trained as an advantage critic for sizing, ranks topologies below chance (Appendix C). Tolerance to process variation and to layout parasitics beyond the SPICE switch model is likewise unaddressed.

7 Conclusion

We presented G-DiffPS, a graph-conditioned conditional flow matching policy that amortizes RF phase-shifter synthesis into a ≈2ms forward pass. Its physics-derived logarithmic action space improves first-pass yield by 465×, while graph conditioning enables zero-shot transfer to unseen topologies. The three claims of Section 1 are supported by the results: C1 by Figure 2, showing zero inference SPICE calls compared with 10^2–10^4 calls for iterative optimizers; C2 by Table 2, demonstrating 99and C3 by Table 5 and Figure 3, where four of six held-out topologies achieve 61–100attributed to frequency-band mismatch rather than encoder capacity (Appendix A). The key contribution is therefore not a better asymptotic reward than iterative optimizers, but removing optimization from the deployment loop.

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A Topology Encoder Diagnostic: GIN Design Rationale and LOOCV Invariance

This appendix proves that the zero-shot ceiling in C3 (Section 5.3) is set by the CFM actor's parameter prior, not by the encoder's representational capacity, and explains why GIN is nonetheless the theoretically correct architectural choice.

Representation collapse. On the six RF graphs (5–10 nodes, sparse one-hot features) a two-layer GraphSAGE encoder with mean aggregation and a post-pool LayerNorm forces all six topology embeddings onto a single hypersphere (mean pairwise $L_2 = 0.062$, cosine similarity >0.999). An earlier stacked-ReLU variant collapsed embeddings to the all-zero vector ($L_2 = 0.000$). Our GIN encoder—*sum* aggregation, `global_add_pool` readout, injected node degree, node-level LayerNorm—restores discriminative geometry: mean pairwise $L_2 = 4.03$, with structurally similar topologies nearest (All Pass ↔ Switched Filter) and dissimilar ones farthest (Vector Modulator ↔ All Pass).

In-distribution performance is encoder-invariant. Table 6 reports zero-shot LOOCV compliance for all three encoders under an otherwise identical framework (CFM actor, log sizing, IQL, 3 seeds). Despite the $0.06 \rightarrow 4.03$ jump in embedding diversity, per-topology compliance is statistically indistinguishable across architectures. The in-distribution gains do not separate the encoders within this six-topology benchmark. The zero-shot ceiling is set by the CFM actor's learned parameter prior—bounded by the 10–28 GHz training distribution—not by the encoder's representational capacity, which is why All Pass (2.4 GHz) fails at 0% regardless of encoder.

Why GIN nonetheless. GIN is provably more expressive than mean-aggregation architectures under the 1-WL graph isomorphism test [14], and sum pooling preserves graph-size information that mean pooling discards. These properties are not exposed by the current six-topology set, where graphs are small and structurally distinct enough that even collapsed encoders produce separable downstream behavior. They become relevant as the topology set grows: a richer benchmark with structurally similar circuits will surface the representational gap the current data cannot. We adopt GIN as the theoretically grounded baseline, with the honest caveat that its advantage is not empirically demonstrated within this benchmark.

Table 6: LOOCV compliance (%) is invariant across encoders; embedding L_2 diversity is not. Same framework throughout; only the topology encoder differs.

Topology	Dying-ReLU	SAGE+LN	GIN (ours)
Loaded_Line	76.5%	75.2%	76.7%
Switched_Line	100.0%	100.0%	100.0%
Reflection_Type	59.2%	58.3%	61.2%
Switched_Filter	17.8%	18.3%	17.7%
Vector_Modulator	99.7%	98.8%	100.0%
All_Pass	0.0%	0.0%	0.0%
Embed. L_2	0.00	0.06	4.03

B Training-Budget Detail: CFM Stability and Convergence Margin

This appendix demonstrates that CFM is the correct generative head choice over DDPM by reporting a catastrophic training collapse in DDPM that never occurred in CFM, and confirms that the 10,000-step budget provides more than a threefold margin past convergence.

Section 5.4 and Figure 4 compare CFM and DDPM under matched conditions (same GIN encoder, log sizing, IQL critic, 3 seeds).

Stability. The operative difference is stability, not convergence speed. DDPM seed 42 suffered a catastrophic reward collapse at step 9,998 (reward −4.78), a failure mode absent across all CFM seeds. In an online loop where reward arrives from noisy SPICE evaluations and occasional simulation divergences, a generative head that can destabilize near the end of training is not acceptable—a single collapse erases the policy learned over the entire run. CFM exhibited no such collapse across any seed or topology.

Convergence budget. Both actors otherwise saturate by ≈2,500 steps. Measured over the final 4,000 steps the smoothed rewards agree within seed-to-seed standard deviation. The 10,000-step budget used throughout the paper therefore carries a threefold margin past convergence, and reported numbers are insensitive to the exact stop point.

Inference determinism. At deployment the gap is decisive: the CFM actor reads a design from a single deterministic ODE trajectory, whereas DDPM must integrate a stochastic reverse process. Reproducibility of the synthesized design is a practical requirement in an iterative EDA flow; CFM satisfies it, DDPM does not.

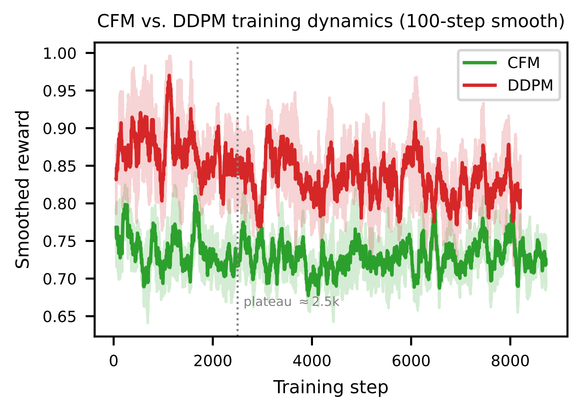

Figure 4: CFM vs. DDPM training reward (100-step smoothing, mean over 3 seeds, GIN encoder). Both saturate near the same plateau by ≈2,500 steps. DDPM seed 42 suffered a catastrophic collapse at step 9,998 (reward −4.78, off the plotted range); CFM showed no collapses across any seed. The dotted line marks the convergence plateau that sets the training budget.

C Topology Recommendation: A Negative Result from V_ψ

This appendix shows that the value head V_ψ, trained as an advantage critic for component sizing, carries no useful topology-selection signal—ranking topologies below random chance—and explains why this outcome is expected and does not affect the validity of the sizing results in the main paper.

G-DiffPS synthesizes parameters for a *given* topology. A natural question is whether the same model can *recommend* a topology for a target spec. We probed this by ranking the six topologies with the value head $V_\psi(\mathbf{s}, \mathbf{z}_{\text{topo}})$ and comparing the top-ranked choice against the SPICE-empirical-best topology over 150 random specifications. The value head ranks the correct topology first only 10% of the time (top-2: 31%), below the 16.7% random baseline. With 150 specifications and 6 classes, 10% top-1 accuracy falls within normal variation around chance; there is no evidence of systematic anti-correlation, only of V_ψ carrying no useful topology-selection signal. This is expected: V_ψ is trained as an advantage critic to weight the sizing loss, not as a topology classifier, and the empirical-best topology is dominated by a few high-ceiling structures (Switched Line, Reflection Type) that the value head cannot distinguish from specification alone. Automatic topology selection is unsolved by the current model and is deferred to future work (Section 7); all results in this paper condition on a specified topology.

D LLM-as-Sizer Baseline: Resonance-Constrained Failure Mode

This appendix establishes the performance ceiling of pretrained LLMs as zero-shot component-value generators, and demonstrates that their failures are structurally tied to the sub-pF resonance constraints that motivate the physics-informed log warp in Section 4—not to prompt quality.

As an additional reference point we evaluate a pretrained large language model as a zero-shot component-value generator, in the spirit of LLM-guided SPICE generation [10] and LLM-driven analog design automation [5, 11]. We prompt Claude Sonnet with a structured specification-to-component-value instruction: given the topology name, target f_c, S_{11}, S_{21}, and phase coverage, generate SPICE-ready component values directly, zero-shot, with no SPICE feedback. Three samples per scenario are drawn; the best-reward design is retained.

The failure pattern is not uniform and is itself informative. On broad-tolerance topologies—Switched Line (f_c = 14, 24 GHz, reward ≈0.80–0.85) and Reflection Type (f_c = 28 GHz, reward 0.82)—Claude Sonnet produces positive-reward designs that clear the spec. On resonance-sensitive mmWave scenarios—Loaded Line at 28 and 38 GHz—it fails outright (reward −5.0, SPICE-rejected). The dividing line is precisely the sub-pF capacitor at the heart of a sharp LC resonance that the physics-informed log warp is designed to resolve (Section 4). The LLM's pretrained prior distributes component values broadly; it has no mechanism to concentrate probability on a sub-pF resonant window, and no SPICE-in-the-loop signal to correct the miss. G-DiffPS's ≥ 99% in-distribution yield against the LLM's partial success rate reflects this structural gap, not a prompt-engineering shortfall.